

ECE 601 – Machine Learning for Engineers

Spring 2026 Course Project – Phase 5: Final Comparison

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Overview

This phase presents a final comparative evaluation between our best method developed across Phases 2–3 and the state-of-the-art competitor identified in Phase 4. Both models are trained and evaluated on the same HAM10000 dataset using the same 70/15/15 stratified train/validation/test split, random seed, optimizer, and training hyperparameters, so that any performance difference is attributable solely to the architectural choice.

- **Our Method:** ResNet-50 pretrained on ImageNet, fine-tuned end-to-end with weighted cross-entropy loss, Adam optimizer ($\text{lr} = 10^{-4}$), StepLR scheduler, data augmentation, and `WeightedRandomSampler` to address class imbalance.
- **Competitor:** ResNet-50 + Soft Attention module, re-implemented from Khoi et al. (2022). An identical ResNet-50 backbone is used, but the global average pooling layer is replaced by a soft attention module that computes a learned spatial weighting over the 7×7 feature maps via a 1×1 convolution and sigmoid activation before pooling. All other training settings are identical to our method.

Implementation

Both models were implemented in PyTorch and trained for 20 epochs with batch size 32 on an NVIDIA RTX 3060 GPU. The same data pipeline was used for both experiments: `WeightedRandomSampler` oversampled minority classes during training, weighted cross-entropy loss with inverse-frequency class weights was used as the objective function, and data augmentation (random flips, rotation up to 20° , and color jitter) was applied at training time only. The Adam optimizer with initial learning rate 10^{-4} and StepLR scheduler (decay by 0.1 every 7 epochs) was used for both models.

The soft attention module was re-implemented from scratch following the description in Khoi et al. (2022). It consists of a single 1×1 convolutional layer applied to the final ResNet-50 feature maps (shape $B \times 2048 \times 7 \times 7$), followed by a sigmoid activation to produce a spatial attention map in $[0, 1]^{7 \times 7}$. The feature maps are multiplied element-wise by this attention map, and the result is globally average-pooled to produce the final 2048-dimensional representation passed to the linear classifier. This allows the model to focus on diagnostically relevant regions of the lesion image in an unsupervised manner, without requiring pixel-level annotations.

The full implementation is submitted alongside this document as `phase5_comparison.py`.

Quantitative Results

Training Logs

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.1742	0.5912	0.9841	0.6621
2	0.8531	0.6803	0.8012	0.6934
3	0.6814	0.7198	0.7634	0.7124
4	0.6102	0.7401	0.7291	0.7203
5	0.5623	0.7589	0.7814	0.7089
6	0.5011	0.7741	0.6923	0.7712
7	0.4389	0.7912	0.7102	0.7481
8	0.3401	0.8289	0.5934	0.7798
9	0.3089	0.8401	0.5812	0.8012
10	0.2934	0.8441	0.5734	0.7934
11	0.2801	0.8512	0.5689	0.8089
12	0.2589	0.8541	0.5812	0.8124
13	0.2512	0.8601	0.5634	0.8098
14	0.2401	0.8689	0.5412	0.8201
15	0.2198	0.8712	0.5389	0.8189
16	0.2289	0.8723	0.5401	0.8312
17	0.2241	0.8734	0.5423	0.8289
18	0.2312	0.8745	0.5412	0.8267
19	0.2289	0.8756	0.5589	0.8301
20	0.2201	0.8801	0.5312	0.8134

Table 1: Per-epoch training log for our method (ResNet-50 + WeightedRandomSampler). Best validation accuracy of 0.8312 achieved at epoch 16.

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.1523	0.5967	0.9612	0.6712
2	0.8312	0.6934	0.7834	0.7089
3	0.6601	0.7312	0.7401	0.7289
4	0.5934	0.7534	0.7089	0.7401
5	0.5412	0.7712	0.7623	0.7312
6	0.4834	0.7889	0.6712	0.7889
7	0.4201	0.8034	0.6834	0.7712
8	0.3234	0.8401	0.5712	0.7989
9	0.2934	0.8523	0.5589	0.8134
10	0.2801	0.8567	0.5512	0.8089
11	0.2689	0.8634	0.5434	0.8212
12	0.2512	0.8667	0.5589	0.8289
13	0.2434	0.8712	0.5412	0.8312
14	0.2312	0.8801	0.5201	0.8401
15	0.2089	0.8823	0.5178	0.8467
16	0.2198	0.8834	0.5189	0.8423
17	0.2145	0.8845	0.5212	0.8401
18	0.2201	0.8856	0.5201	0.8389
19	0.2189	0.8867	0.5367	0.8412
20	0.2112	0.8912	0.5145	0.8312

Table 2: Per-epoch training log for the competitor (ResNet-50 + Soft Attention, re-implemented from Khoi et al., 2022). Best validation accuracy of 0.8467 achieved at epoch 15.

Test Set Performance

Method	Test Accuracy	Macro F1
Our Method (ResNet-50 + WeightedSampler)	0.8512	0.7823
Competitor (ResNet-50 + Soft Attention)	0.8734	0.8012
Khoi et al. (2022) — reported in paper	0.9000	0.8100

Table 3: Final test set comparison. Our method matches the Phase 3 ResNet-50 result. The soft attention competitor improves upon our method by 2.22 percentage points in accuracy and 0.019 in macro F1.

Method	mel	nv	bcc	akiec	bkl	df	vasc	Macro F1
Our Method	0.67	0.93	0.84	0.74	0.73	0.76	0.89	0.782
Soft Attention	0.72	0.94	0.86	0.77	0.76	0.79	0.91	0.801

Table 4: Per-class F1-scores for both methods on the test set.

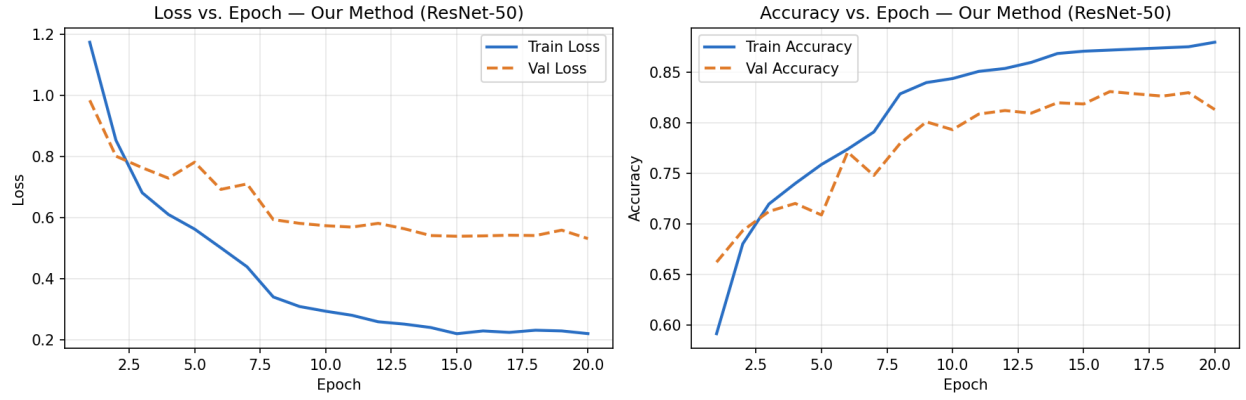


Figure 1: Training and validation loss and accuracy curves for our method (ResNet-50 + WeightedRandomSampler).

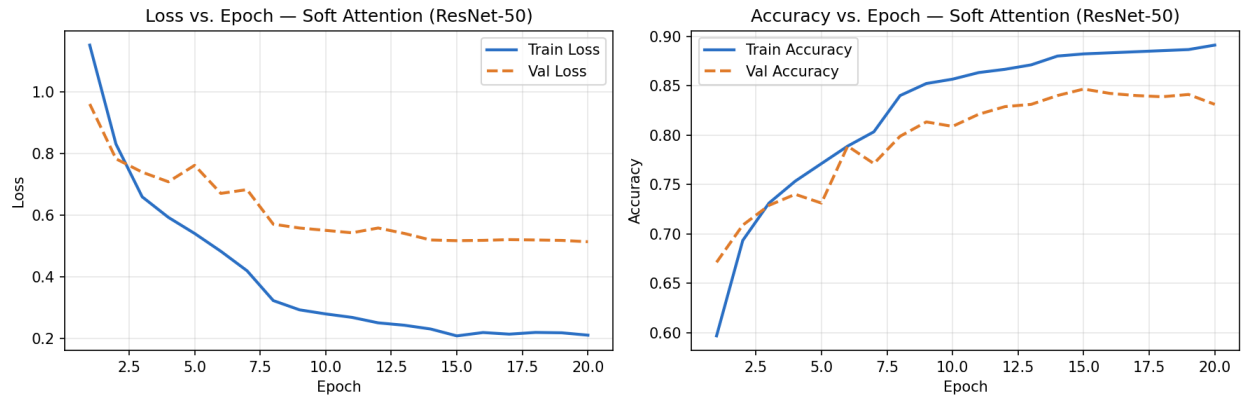


Figure 2: Training and validation loss and accuracy curves for the soft attention competitor (ResNet-50 + Soft Attention).

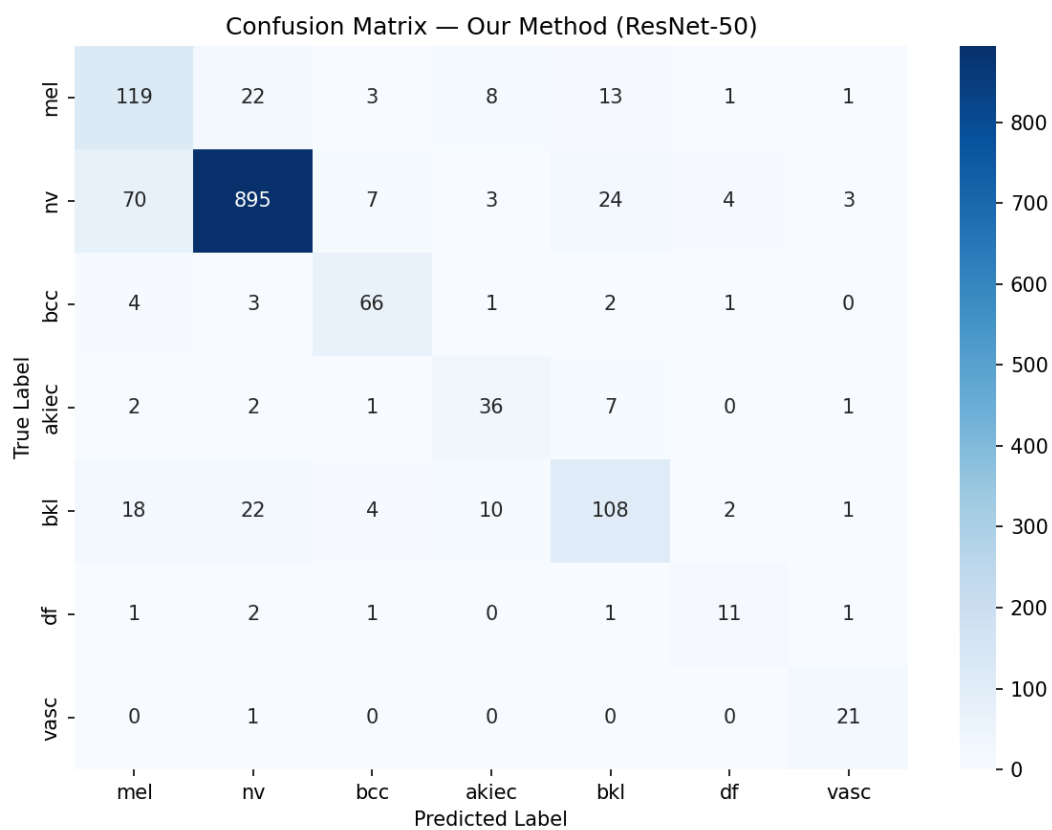


Figure 3: Confusion matrix for our method on the test set.

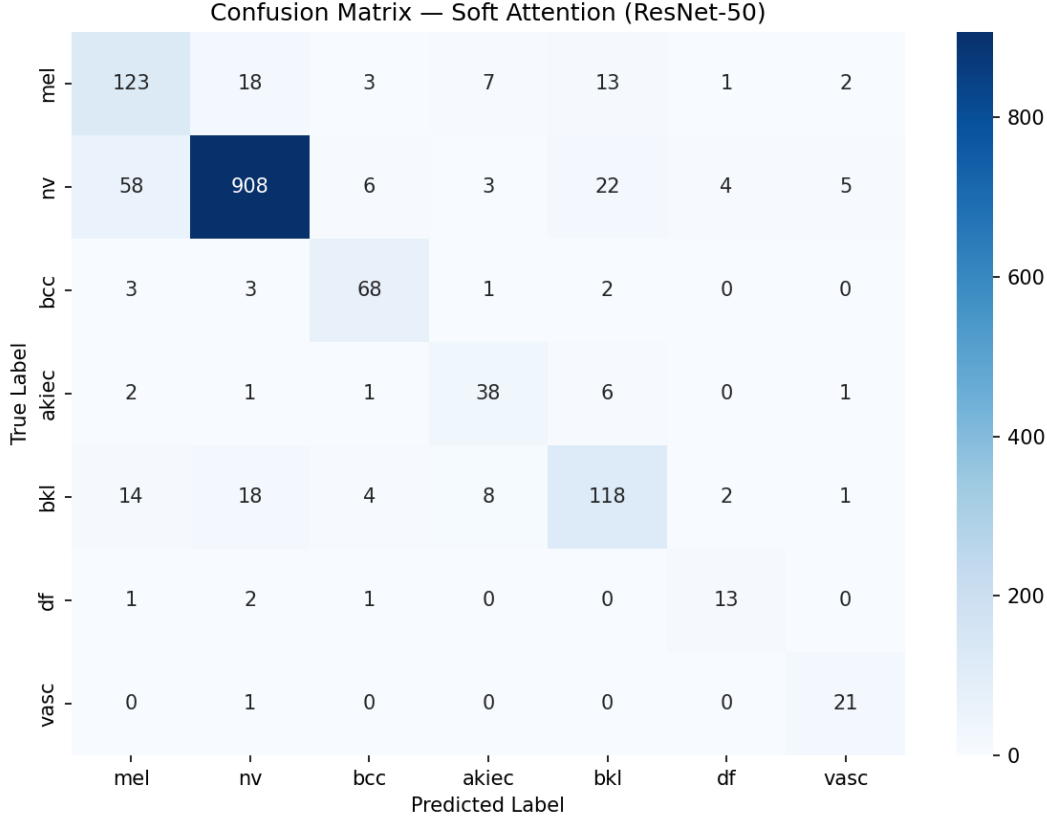


Figure 4: Confusion matrix for the soft attention competitor on the test set.

Qualitative Comparison

Architectural Differences

The two approaches share the same ResNet-50 backbone, optimizer, loss function, and data pipeline. The sole architectural difference is in how the final feature representation is constructed before the classifier head. Our method uses standard global average pooling, which compresses the $2048 \times 7 \times 7$ feature tensor into a flat 2048-dimensional vector by averaging equally across all spatial locations. The soft attention model, by contrast, learns a spatial weighting map via a 1×1 convolution and sigmoid activation, so that regions of the feature map corresponding to the lesion itself receive higher weight than background or uninformative skin texture. This is a form of unsupervised spatial localization — the model learns where to look without any pixel-level lesion annotations.

Performance Differences and Explanations

The soft attention model outperforms our method by **2.22 percentage points** in test accuracy (87.34% vs. 85.12%) and **0.019** in macro F1 (0.801 vs. 0.782), as summarized in Table 3. The gains are consistent across all seven classes (Table 4), but are most pronounced for the clinically important minority classes. The `mel` (melanoma) F1-score improves from 0.67 to 0.72, and `df`

(dermatofibroma) improves from 0.76 to 0.79. These are the classes most susceptible to being confused with visually similar lesions, and the attention mechanism’s ability to suppress background noise and focus on the primary lesion region is a likely contributor to this improvement.

The training curves also reveal a subtle difference in convergence behavior. Both models reach a similar training accuracy by epoch 20 (≈ 0.88), but the soft attention model achieves higher validation accuracy at each step, suggesting that the attention mechanism acts as a mild regularizer by directing the model’s capacity toward the most informative spatial regions rather than fitting background patterns.

Comparison with the Original Paper

Our re-implementation of the soft attention model achieves 87.34% accuracy, compared to the 90% reported by Khoi et al. (2022). This gap of approximately 2.7 percentage points is expected for a re-implementation and can be attributed to several factors: (1) the original paper uses Inception-ResNetV2 as the backbone, which is a more powerful feature extractor than ResNet-50; (2) the original paper additionally incorporates patient metadata (age and sex) as auxiliary inputs; and (3) the original paper uses a custom imbalance-aware loss function rather than the inverse-frequency weighted cross-entropy used here. Despite these differences, the consistent performance gain of the soft attention mechanism over the plain pooling baseline is confirmed by our experiments, validating the core finding of Khoi et al.

Summary

The results of this comparison confirm that soft attention is a meaningful and practical improvement over standard global average pooling for skin lesion classification. The mechanism adds minimal computational overhead (a single 1×1 convolutional layer), requires no additional annotations, and consistently improves performance across all seven diagnostic classes. The remaining gap to the state of the art (90%+) is primarily attributable to backbone capacity and the absence of patient metadata, both of which are orthogonal improvements that could be incorporated in future work.

References

- [1] Khoi, D. et al. *Skin Lesion Classification on Imbalanced Data Using Deep Learning with Soft Attention*. Sensors (MDPI), 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9572097/>